



Girraween High School

2018 Year 12 Mathematics Half Yearly Examination

General Instructions

- Working Time - 1 hour & 30 minutes
- Calculators and ruler may be used
- All necessary working out must be shown
- Write on one side of the paper only

Total Marks - 98

- Attempt all questions
- Marks may be deducted for careless or badly arranged work
- Start each question on a new sheet of paper

Question 1 (1 mark)

What is the value of $\frac{\pi^2}{6}$ correct to 3 significant figures?

- A. 1.64 B. 1.65 C. 1.644 D. 1.645

Question 2 (1 mark)

Which one of the following statements is true for the equation $7x^2 - 5x + 2 = 0$

- A. It has no real roots
B. It has one real root
C. It has two distinct roots
D. It has three real roots

Question 3 (1 mark)

For what values of x is the curve $f(x) = 2x^3 + x^2$ concave down?

- A. $x < -\frac{1}{6}$ B. $x > -\frac{1}{6}$ C. $x < -6$ D. $x > 6$

Question 4 (1 mark)

What is the value of $\lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1}$?

- A. 2 B. 1 C. undefined D. 0

Question 5 (1 mark)

The value of $\sum_{k=3}^{10} 2k + 1$ is

- A. 120 B. 91 C. 122 D. 112

Question 6 (1 mark)

The quadratic equation $x^2 + 3x - 1 = 0$ has roots α and β .

What is the value of $\alpha\beta + (\alpha + \beta)$?

- A. 4 B. 2 C. -4 D. -2

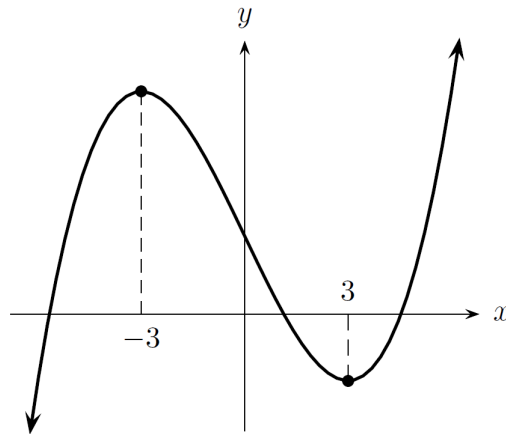
Question 7 (1 mark)

The angle of inclination that the line $3x + 2y - 7 = 0$ makes with the positive x -axis is closest to:

- A. 56° B. 124° C. 34° D. 146°

Question 8 (1 mark)

For the graph of $y = f(x)$, for what x values is $f'(x)$ negative?



- A. $x < -3$ and $x > 3$ B. $-3 < x < 3$ C. $x \leq -3$ and $x \geq 3$ D. $-3 \leq x \leq 3$

Question 9 (1 mark)

The domain of the function $f(x) = \frac{1}{\sqrt{4x^2 - 1}}$ is:

- A. $-\frac{1}{2} < x < \frac{1}{2}$
B. $x < -\frac{1}{2}$ and $x > \frac{1}{2}$
C. $-\frac{1}{2} \leq x \leq \frac{1}{2}$
D. $x \leq -\frac{1}{2}$ and $x \geq \frac{1}{2}$

Question 10 (1 mark)

If y is decreasing at an increasing rate, which of the following is true?

- A. $\frac{dy}{dx} < 0$ and $\frac{d^2y}{dx^2} < 0$
B. $\frac{dy}{dx} > 0$ and $\frac{d^2y}{dx^2} < 0$
C. $\frac{dy}{dx} < 0$ and $\frac{d^2y}{dx^2} > 0$
D. $\frac{dy}{dx} > 0$ and $\frac{d^2y}{dx^2} > 0$

Question 11 (20 marks)

(a) Find the derivative of the following:

i. $y = e^{2x+1} - e^{-\frac{1}{2}x}$ [2]

ii. $y = \frac{1}{1 + 2e^x}$ [2]

iii. $y = \frac{x^2 + 1}{x - 1}$ [2]

iv. $y = 3x(2x + 1)^3$ [2]

v. $y = \frac{2 + e^x}{e^{2x}}$ [2]

(b) Find the primitive of the following:

i. $\frac{e^{2x}}{2}$ [2]

ii. $\sqrt{2x - 1}$ [2]

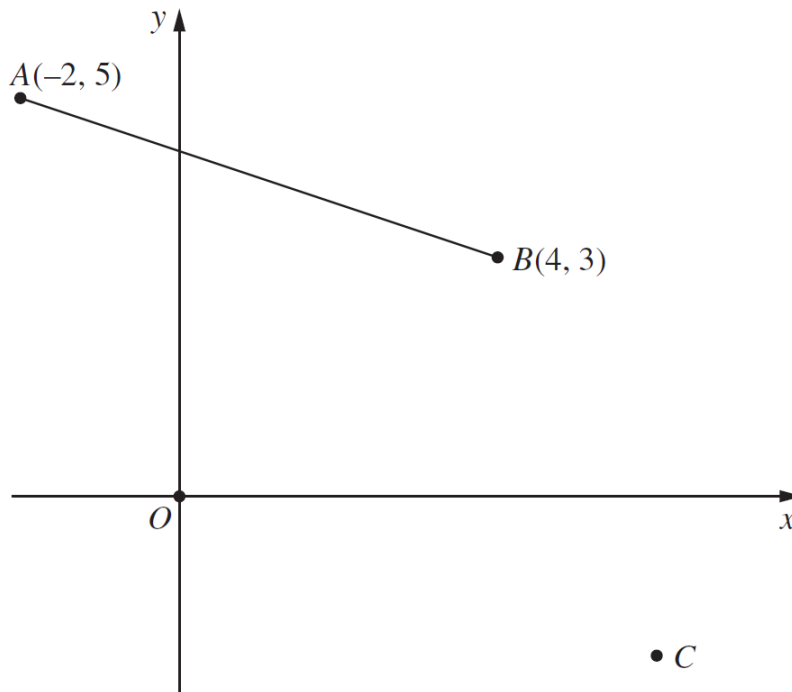
(c) Find the values of a and b such that $\frac{1}{4 - \sqrt{13}} = a + b\sqrt{13}$, where a and b are rational numbers. [3]

(d) Prove that $\sin \theta + \cot \theta \cos \theta = \operatorname{cosec} \theta$ [3]

The exam continues on the next page

Question 12 (19 marks)

- (a) The diagram below shows the points $A(-2, 5)$, $B(4, 3)$ and $O(0, 0)$. The point C is the fourth vertex of the parallelogram $OABC$.



- i. Show that the equation of AB is $x + 3y - 13 = 0$. [2]
 - ii. Show that the length of AB is $2\sqrt{10}$. [1]
 - iii. Calculate the perpendicular distance from O to the line AB . [1]
 - iv. Calculate the area of parallelogram $OABC$. [1]
 - v. Hence or otherwise find the perpendicular distance from O to the line BC . [2]
- (b) The second term of arithmetic progression is 7 and the seventh term is 52
- i. Find the common difference. [2]
 - ii. Find the value of the first term which is greater than 1000 [2]
 - iii. Find the sum of the first ten terms. [1]
- (c) The equation $(x - 1)^2 = -4y + 16$ represents a parabola.
- i. Find the coordinates of the vertex. [2]
 - ii. Find the focal length and the equation of directrix. [2]
 - iii. Sketch parabola, clearly showing the directrix, the focus and the x -intercept. [3]

The exam continues on the next page

Question 13 (18 marks)

- (a) Michael buys five tickets in a raffle in which 20 tickets are sold. Three different tickets are to be drawn out without replacement for first, second and third prizes. Find the probability that:

- i. Michael wins all three prizes. [2]
- ii. Michael does not win a prize. [2]
- iii. Michael wins at least one prize. [1]
- iv. Michael wins exactly one prize. [2]

- (b) i. Find $\frac{d}{dx} e^{1-x^2}$ [1]

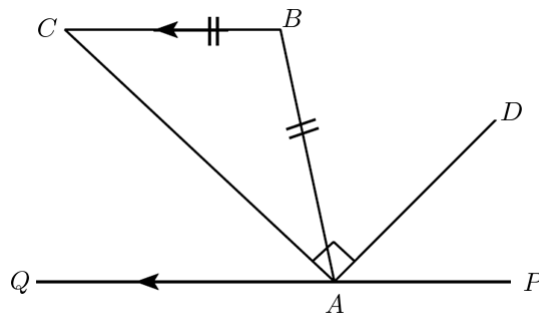
- ii. Hence or otherwise find $\int x e^{1-x^2} dx$ [2]

- (c) i. Use the trapezoidal rule with 5 function values to find an approximation to [3]

$$\int_1^5 \frac{1}{x} dx.$$

- ii. State whether the approximation found in part (i) is greater or less than the exact value of $\int_1^5 \frac{1}{x} dx$. Justify your answer. [1]

- (d) In the diagram below ABC is a triangle with $AB = BC$. The line PQ passes through A parallel to BC , and the line AD is perpendicular to AC .



- i. Let $\angle BAC = \theta$ and prove that AC bisects $\angle QAB$. [2]
- ii. Prove that AD bisects $\angle PAB$. [2]

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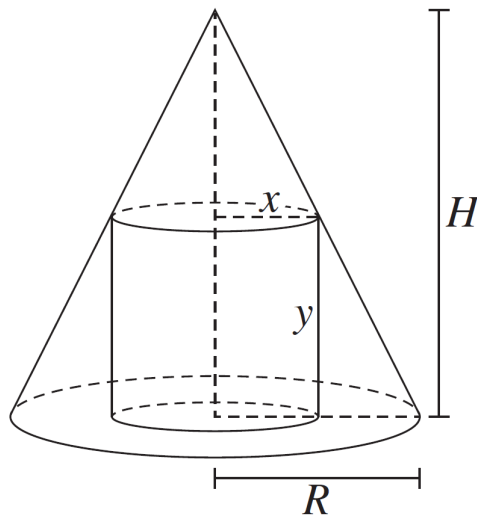
Question 14 (17 marks)

- (a) Consider the function $f(x) = xe^x$
- i. Find the coordinates of the stationary points and determine their nature. [3]
 - ii. Find the coordinates of the points of inflexion. [2]
 - iii. State the behaviour of $y = f(x)$ as $x \rightarrow \infty$ and as $x \rightarrow -\infty$. [2]
 - iv. Sketch the graph of the curve $y = f(x)$, showing intercepts, stationary points and point of inflexion. [3]
- (b) Let A and B be fixed points $(-1, 0)$ and $(1, 2)$. Let P be the variable point (x, y) such that $\angle APB = 90^\circ$. Find the equation of the locus of P . [3]
- (c) Consider the series $1 + x + (1 + x)^2 + (1 + x)^3 + \dots$
- i. Show that the series is geometric. [1]
 - ii. Find the values of x for which this series has a limiting sum. [2]
 - iii. Find the limiting sum of this series in terms of x . [1]

The exam continues on the next page

Question 15 (14 marks)

- (a) Find the volume of the solid generated by rotating the area enclosed by the curve $y = x^2 + 1$, $x = 1$ and the coordinate axes around the y -axis. [3]
- (b) The graph of $y = e^{kx}$ has a tangent at $x = a$ that passes through the origin, where k and a are constants.
- i. Show that $ka = 1$. [3]
- ii. Find the area bounded by $y = e^{x/2}$ and its tangent at $x = 2$ and the y -axis. [3]
- (c) The diagram below shows a cylinder of radius x and height y inscribed in a cone of radius R and height H , where R and H are constants.



The volume of a cone of radius r and height h is $\frac{1}{3}\pi r^2 h$.

The volume of a cylinder of radius r and height h is $\pi r^2 h$.

- i. Show that the volume, V , of the cylinder can be written as [2]

$$V = \frac{H}{R}\pi x^2(R - x)$$

- ii. Find the value of x that maximises V . [3]

End of exam

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Q1 (A)

Q2 $\Delta = 25 - 4(7)(2) < 0$

$\therefore$ (A)

Q3 $f(x) = 2x^3 + x^2$

$f'(x) = 6x^2 + 2x$

$f''(x) = 12x + 2$

$12x + 2 < 0$

$12x < -2$

$x < -\frac{1}{6} \therefore$ (A)

Q4 $\lim_{x \rightarrow 1} \frac{(x-1)(x+1)}{x-1} = 2 \therefore$ (A)

Q5 $a = 7 \quad d = 2 \quad n = 8$

$S_n = \frac{8}{2} (7 + 21) = 112 \therefore$ (D)

Q6 $\alpha\beta = -1 \quad \alpha + \beta = -3 \therefore$ (C)

Q7 $3x + 2y - 7 = 0$

$2y = -3x + 7$

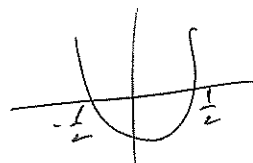
$y = -\frac{3}{2}x + \frac{7}{2}$

$\tan \alpha = -\frac{3}{2} \quad \alpha = 123^\circ 41' \therefore$ (B)

Q8 (B)

Q9 $4x^2 - 1 > 0$

$(2x-1)(2x+1) > 0$

 $\therefore x > \frac{1}{2}$
& $x < -\frac{1}{2}$

$\therefore$ (B)

Q10 (C) or (A)

Q11

(a) $y = e^{2x+1} - e^{-\frac{1}{2}x}$

$y' = 2e^{2x+1} + \frac{1}{2}e^{-\frac{1}{2}x}$

$\therefore y = (1+2e^x)^{-1}$

$y' = -(1+2e^x)^{-2} \times 2e^x$

$y' = \frac{-2e^x}{(1+2e^x)^2}$

iii. $u = x^2 + 1 \quad v = x - 1 \quad y' = \frac{2x(x-1) - (x^2+1)}{(x-1)^2}$
 $u' = 2x \quad v' = 1$

$y' = \frac{x^2 - 2x - 1}{(x-1)^2}$

iv. $u = 3x \quad v = (2x+1)^3$

$u' = 3 \quad v' = 3(2x+1)^2 \times 2$

$v' = 6(2x+1)^2$

$y' = 3(2x+1)^3 + 18x(2x+1)^2$

Q11

(a) v. $y = 2e^{-2x} + e^{-x}$
 $y' = -4e^{-2x} - e^{-x}$

(b) (i) $\frac{1}{2} \int e^{x/2} dx$
 $= e^{x/2} + C$

(ii) $\int (2x-1)^{1/2} dx$
 $= \frac{(2x-1)^{3/2}}{2 \times 3/2} + C$
 $= \frac{1}{3} (2x-1)^{3/2} + C$

(c) $\frac{1}{4-\sqrt{13}} \times \frac{4+\sqrt{13}}{4+\sqrt{13}}$
 $= \frac{4+\sqrt{13}}{16-13} = \frac{4}{3} + \frac{1}{3}\sqrt{13}$
 $\therefore a = \frac{4}{3} \text{ \& } b = \frac{1}{3}$

(d) LHS $= \sin \theta + \cot \theta \cos \theta$
 $= \sin \theta + \frac{\cos^2 \theta}{\sin \theta}$
 $= \frac{\sin^2 \theta + \cos^2 \theta}{\sin \theta}$
 $= \frac{1}{\sin \theta} = \operatorname{cosec} \theta = \text{RHS}$

Q12

(a) (i) $m = \frac{5-3}{-2-4} = \frac{2}{-6} = -\frac{1}{3}$

$y-3 = -\frac{1}{3}(x-4)$

$3y-9 = -x+4$

$x+3y-13=0$

(ii) $D_{AB} = \sqrt{6^2 + (-2)^2}$
 $= \sqrt{40} = 2\sqrt{10}$

(iii) $d = \frac{|-13|}{\sqrt{1^2+3^2}} = \frac{13}{\sqrt{10}}$

(iv) $A = 2\sqrt{10} \times \frac{13}{\sqrt{10}} = 26 \text{ u}^2$

(v) $D_{OK} = \sqrt{2^2 + 5^2}$
 $= \sqrt{29}$

$\therefore \sqrt{29} \times d = 26$

$\therefore d = \frac{26}{\sqrt{29}}$

(b)

(i) $a+d=7$

$a+6d=52$

$\therefore 5d=45 \therefore d=9$

Q12

(b)

$$(i) a = -2$$

$$a + (n-1)d > 1000$$

$$-2 + (n-1) \times 9 > 1000$$

$$-2 + 9(n-1) > 1000$$

$$9n - 9 > 1002$$

$$9n > 1011$$

$$n > \frac{1011}{9} \therefore n = 113$$

$$T_{113} = -2 + 112 \times 9 = 1006$$

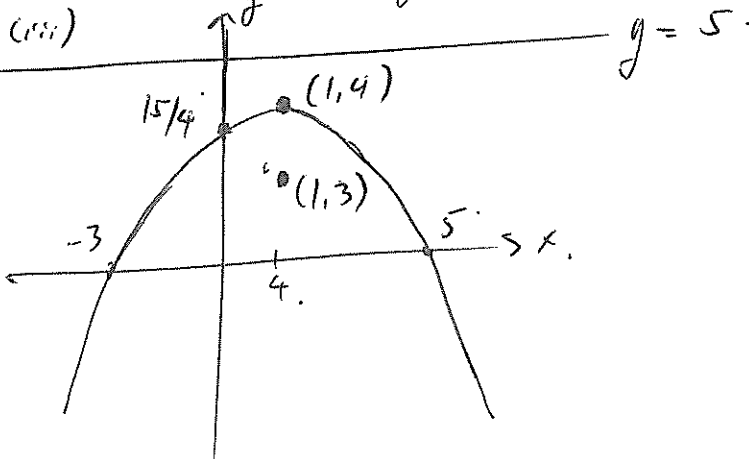
$$(ii) S_{113} = \frac{113}{2} (2(-2) + 9 \times 9) = 385$$

(c)

$$(i) (x-1)^2 = -4(y-4)$$

$$V = (1, 4)$$

$$(ii) a = 1 \quad D: y = 5$$



Q13

$$(i) \frac{5}{20} \times \frac{9}{19} \times \frac{3}{18} = \frac{1}{114}$$

$$(ii) \frac{15}{20} \times \frac{14}{19} \times \frac{13}{18} = \frac{91}{228}$$

$$(iii) 1 - \frac{91}{228} = \frac{137}{228}$$

$$(iv) 3 \times \frac{5}{20} \times \frac{15}{19} \times \frac{14}{18} = \frac{35}{76}$$

(b)

$$(i) \frac{d}{dx} e^{1-x^2} = -2x e^{1-x^2}$$

$$(ii) \int -2x e^{1-x^2} dx = e^{1-x^2}$$

$$\therefore \int x e^{1-x^2} dx = -\frac{1}{2} e^{1-x^2} + C$$

(c)

$$(i) h = \frac{5-1}{4} = 1$$

$$A \approx \frac{1}{2} \left[\frac{1}{1} + 2\left(\frac{1}{2}\right) + 2\left(\frac{1}{3}\right) + 2\left(\frac{1}{4}\right) + \left(\frac{1}{5}\right) \right]$$

$$\approx \frac{101}{60}$$

(ii) Greater, as $y = \frac{1}{x}$ is concave up for $1 \leq x \leq 5$.

(d)

$\angle CAQ = \theta$ (alternate $\angle$'s,
 $CB \parallel QA$).

$$(11) \quad \theta + 90 + \angle DAP = 180 \text{ (Straight L)}$$
$$\angle DAB = 90 - \theta \text{ (adjacent } \angle)$$

cx 14

(i) $n' = 1$ $v' = e^n$

$$f'(x) = e^x + x e^x$$

$$f'(x) = e^x (1+x).$$

$$f'(x) = 0 \text{ when } x = -1.$$

$f(-1) = -\frac{1}{e} \therefore$ start pt at $(-1, -\frac{1}{e})$

$$= 2e^x + xe^x.$$

$$f''(-1) = 2e^{-1} - e^{-1}$$

$$= e^{-1} > 0$$

$\therefore$ mark at $(-1, -\frac{1}{e})$

(ii) $f''(x) = 0$ when $e^x(2+x) = 0$.

$$\therefore x = -2$$

$$f(-2) = -2e^{-2} = -\frac{2}{e^2}$$

possible point of inflexion at

$$\left(-2, -\frac{2}{e^2}\right).$$

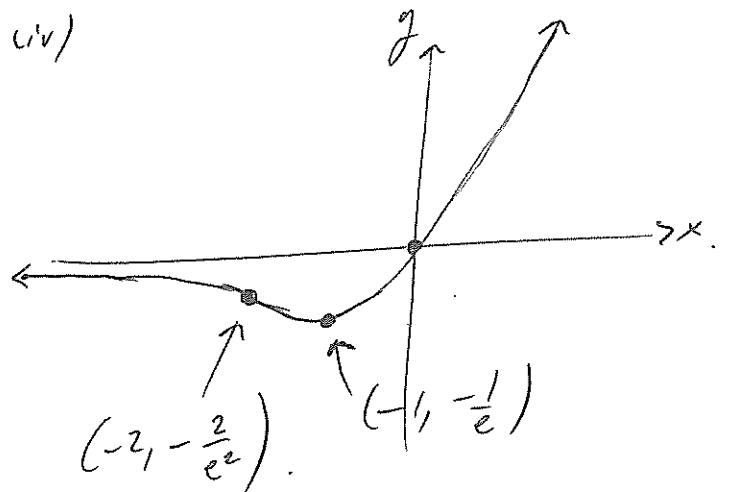
$$\begin{array}{c|c|c|c} x & -3 & -2 & -1 \\ \hline y'' & -\frac{1}{e^3} & 0 & \frac{1}{e} \end{array}$$

$\therefore (-2, -\frac{2}{e^2})$ is a point of inflexion.

(iii) $x \rightarrow \infty$ $y \rightarrow \infty$

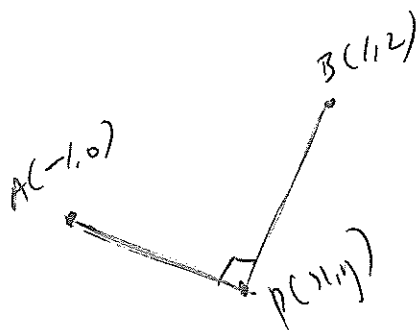
$$x \rightarrow -\infty \quad y \rightarrow 0.$$

(iv)



Q14

(b)



$$\frac{y}{x+1} \times \frac{y-2}{x-1} = -1$$

$$y(y-2) = -(x^2-1)$$

$$y^2 - 2y = -x^2 + 1$$

$$x^2 + y^2 - 2y - 1 = 0$$

(c)

$$(i) \frac{T_2}{T_1} = \frac{(1+n)^2}{(1+n)} = 1+n$$

$$\frac{T_3}{T_2} = \frac{(1+n)^3}{(1+n)^2} = 1+n$$

$\therefore$ Series is geometric.

$$(ii) \text{ Need } -1 < 1+n < 1$$

$$\therefore -2 < n < 0$$

(iii)

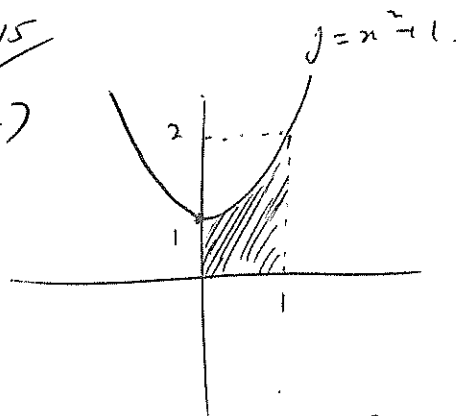
$$S_{\infty} = \frac{a}{1-r}$$

$$= \frac{1+n}{1-(1+n)} = \frac{1+n}{-n}$$

$$= \frac{-1-n}{n}$$

Q15

(a)



$$V = \pi (1)^2 \times 2 - \pi \int_{-1}^1 (y-1) dy$$

$$V = 2\pi - \pi \left[\frac{y^2}{2} - y \right]_{-1}^1$$

$$V = 2\pi - \pi \left(2 - 2 - \left(\frac{1}{2} - 1 \right) \right)$$

$$V = 2\pi - \frac{\pi}{2} = \frac{3}{2}\pi$$

$$(b) (i) y = e^{kx}$$

$$y' = ke^{kx}$$

$$y(a) = e^{ka} \quad y'(a) = ke^{ka}$$

$$(n, e^{ka})$$

$$y - e^{ka} = ke^{ka}(x-a)$$

$$y - e^{ka} = ke^{ka}n - ake^{ka}$$

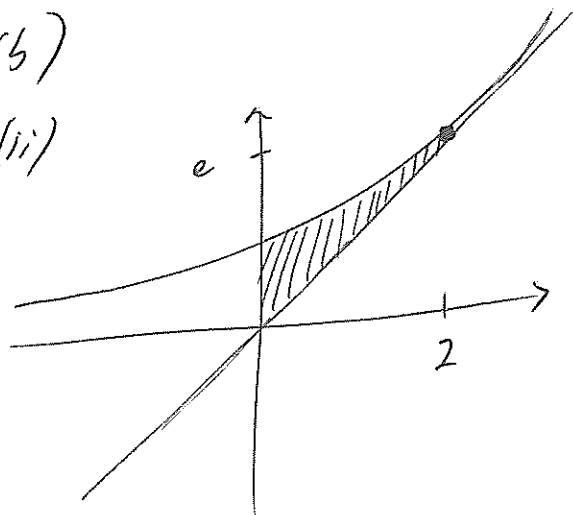
$$y = ke^{ka}n + (1-ak)e^{ka}$$

$$\therefore 1-ak = 0 \quad \therefore ak = 1$$

Q15

(b)

(ii)



$$\begin{aligned}
 A &= \int_0^2 e^{x/2} dx - \frac{1}{2}(2)(e) \\
 &= 2 \left[e^{x/2} \right]_0^2 - e \\
 &= 2(e-1) - e \\
 &= 2e - 2 - e = e - 2.
 \end{aligned}$$

(c)

(i) $\frac{x}{R} = \frac{H-y}{H}$ (ratio of matching sides of $\triangle$'s).

$$\therefore Hx = R(H-y).$$

$$Hx = RH - Ry.$$

$$Ry = RH - Hx.$$

$$y = \frac{H}{R}(R-x).$$

$$V = \pi x^2 y$$

$$\therefore V = \pi x^2 \times \frac{H}{R}(R-x).$$

$$V = \frac{H}{R} \pi x^2 (R-x)$$

$$(ii) V = H \pi x^2 - \frac{H}{R} \pi x^3$$

$$V' = 2H\pi x - \frac{3H}{R} \pi x^2.$$

$$V' = H\pi x \left(2 - \frac{3}{R} x \right).$$

$$\therefore V' = 0 \text{ when } x = 0$$

$$\text{or } 2 - \frac{3x}{R} = 0.$$

$$2 = \frac{3x}{R}$$

$$\therefore x = \frac{2R}{3}.$$

$$V'' = 2H\pi - \frac{6H}{R} \pi x.$$

$$V''\left(\frac{2R}{3}\right) = 2H\pi - \frac{6H}{R} \pi \times \frac{2R}{3}.$$

$$= 2H\pi - 4H\pi < 0$$

$$\therefore x = \frac{2R}{3} \text{ gives max } V.$$